

Exploratory Analysis of Diabetes Hospital Readmissions

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1. Dataset

I chose the Hospital Readmissions dataset from the UCI Machine Learning Repository. It contains diabetes-related hospital encounters from 130 U.S. hospitals between 1999 and 2008, including demographics, admission context, diagnoses, procedures, medications, lab results, and whether the patient was readmitted.

2. Frame the Problem Before Touching the Data

Imaginary client

My imaginary client is the Chief Medical Officer of a regional hospital network that is trying to reduce avoidable 30-day readmissions among patients with diabetes. The network is under pressure because readmissions are costly for patients and hospitals: they signal unresolved clinical complexity, they consume bed capacity, and they can affect reimbursement and quality ratings. The CMO does not need a black-box prediction model first; she needs a clear map of where readmission risk is concentrated so that clinical operations can decide which patient groups deserve earlier discharge planning, medication reconciliation, or post-discharge follow-up.

The decision facing the client is how to allocate limited transitional-care resources. A nurse call, pharmacy review, or rapid follow-up appointment cannot be given with equal intensity to every encounter. The analysis should therefore identify practical risk markers available during or immediately after the hospital stay: age, prior utilization, admission type, discharge destination, medication changes, lab testing, and length of stay.

Three research questions

1. **Which patient groups should receive the highest priority for post-discharge follow-up?** Age, prior healthcare utilization, and discharge destination are likely to identify patient groups with a higher likelihood of near-term readmission.
2. **Should pharmacy review resources be focused on patients who experience medication changes during hospitalization?** The relationship between medication changes, diabetes medications, A1C testing, and 30-day readmission may indicate whether targeted pharmacy review could help reduce avoidable readmissions.
3. **Which hospitalized diabetes patients should automatically receive transitional-care planning before discharge?** Length of stay, number of diagnoses, medications, procedures, admission source, and admission type may indicate greater clinical complexity and a higher likelihood of readmission after discharge.

Prior expectations

Before loading the dataset, I expected older patients to have higher 30-day readmission rates because diabetes complications accumulate with age and comorbidity. I expected patients with more prior inpatient or emergency visits to be substantially more likely to return soon, since prior utilization is often a marker of unstable health and access barriers. I also expected medication changes and abnormal A1C results to be associated with higher readmission risk, but I did not expect them to be clean causal signals because clinicians change medication in response to severity.

3. First Contact with the Data

Data passport

The dataset has 101,766 rows and 50 original columns. I converted ? strings to missing values and added derived fields for 30-day readmission, any readmission, age midpoint, total prior visits, and grouped discharge labels. Overall, 11.2% of encounters were readmitted within 30 days and 46.1% had any recorded readmission.

I treated identifier-like clinical codes such as admission_type_id, discharge_disposition_id, and admission_source_id as categorical variables, not numeric measurements. This avoids reporting meaningless means for codes and instead reports the number of observed categories and the most common codes.

Numeric variables most relevant to the analysis:

variable	missing_rate	min	max	mean	median	std
time_in_hospital	0	1	14	4.396	4	2.985
num_lab_procedures	0	1	132	43.096	44	19.674
num_procedures	0	0	6	1.34	1	1.706
num_medications	0	1	81	16.022	15	8.128
number_outpatient	0	0	42	0.369	0	1.267
number_emergency	0	0	76	0.198	0	0.93
number_inpatient	0	0	21	0.636	0	1.263
number_diagnoses	0	1	16	7.423	8	1.934

Selected categorical and code variables:

variable	missing_rate	n_distinct	top_3
race	0.022	5	Caucasian: 76099; AfricanAmerican: 19210; Hispanic: 2037
gender	0	3	Female: 54708; Male: 47055; Unknown/Invalid: 3
age	0	10	[70-80): 26068; [60-70): 22483; [50-60): 17256
admission_type_id	0	8	1: 53990; 3: 18869; 2: 18480
discharge_disposition_id	0	26	1: 60234; 3: 13954; 6: 12902
admission_source_id	0	17	7: 57494; 1: 29565; 17: 6781
payer_code	0.396	17	MC: 32439; HM: 6274; SP: 5007
medical_specialty	0.491	72	InternalMedicine: 14635; Emergency/Trauma: 7565; Family/GeneralPractice: 7440
readmitted	0	3	NO: 54864; >30: 35545; <30: 11357

The most striking data-quality issue is that weight is missing for almost every encounter, so it cannot support a serious analysis of obesity or dose-related medication questions. payer_code and medical_specialty also have substantial missingness, which limits conclusions about insurance and specialty service lines. Some clinically meaningful variables are encoded as categories rather than measurements: age is binned, diagnoses are ICD-like strings, and lab fields such as A1C are only recorded as thresholds. Finally, the unit of observation is an encounter, not a unique person, so high-utilization patients can appear multiple times and influence encounter-level patterns.

Variation within variables

Most encounters were not followed by a recorded readmission, but the 30-day category is still large enough to be operationally important. The client should not interpret the task as a rare-event problem in which only a tiny fraction of patients matter; more than one in ten encounters returned within 30 days.

The encounters are concentrated among older adults, especially ages 60 to 80. This means an intervention designed around diabetes readmissions will mostly be an intervention for older patients, even before any risk stratification. It also means patterns in the youngest groups should be interpreted cautiously because they represent a much smaller share of encounters.

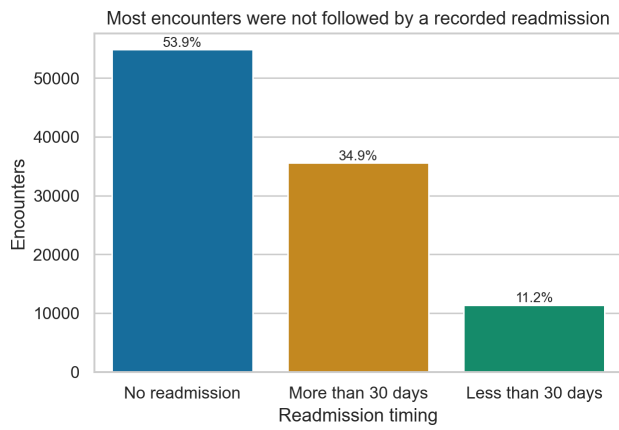


Figure 1: Readmission timing distribution

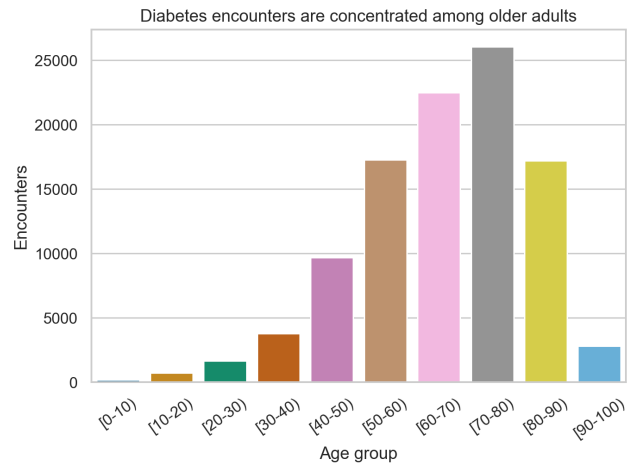


Figure 2: Age distribution

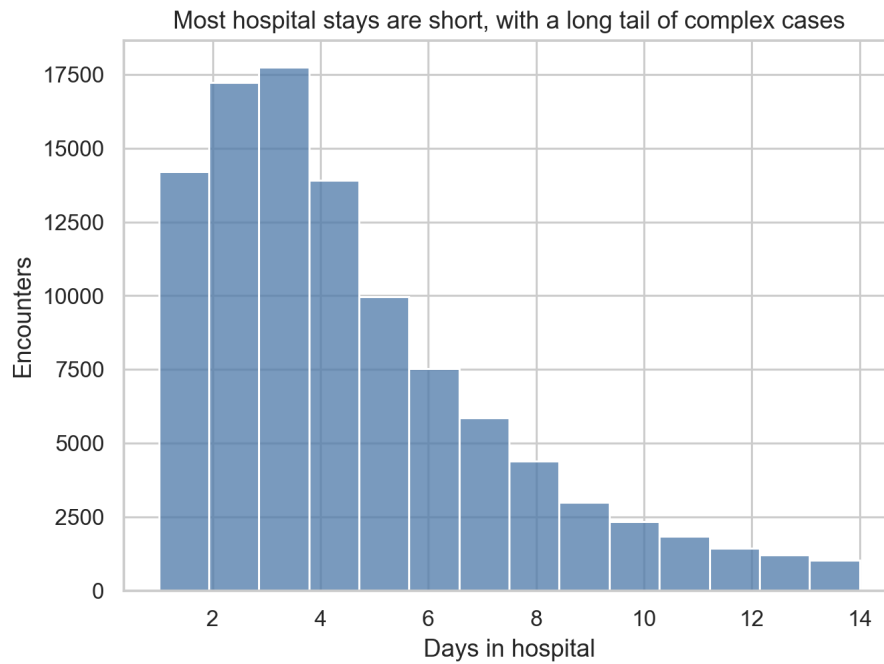


Figure 3: Length of stay distribution

This distribution was consistent with my prior expectation that 30-day readmissions would represent a meaningful but minority share of encounters. The concentration of encounters in ages 60–80 confirmed my prior that older patients would dominate this dataset, though the extent to which the youngest groups are underrepresented was more pronounced than anticipated. Most stays are short, but there is a long right tail up to two weeks. In hospital operations, that tail likely represents more complex cases that consume disproportionate resources. The skew is not just a statistical shape: it suggests that a simple average length of stay would hide the operational difference between routine and complicated admissions. The right skew was expected given the clinical reality of diabetes complications, though the length of the tail suggested a more severe complexity gradient than my prior implied.

Covariation Between Variables

The following three plots are each tied directly to one of the research questions stated above. They examine relationships between patient or encounter characteristics and 30-day readmission, with color and axes used to communicate the relevant grouping or risk gradient.

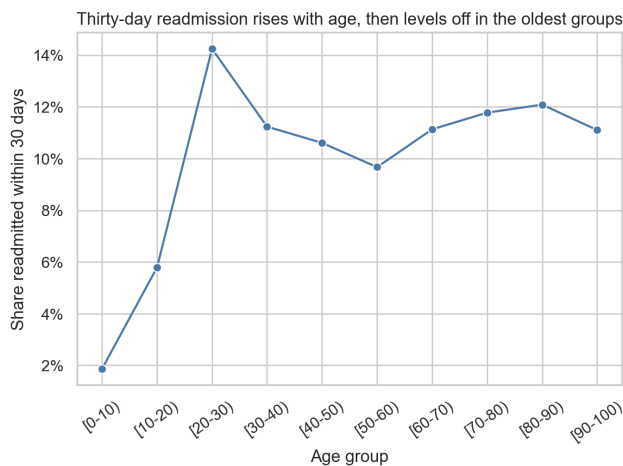


Figure 4: Age group and 30-day readmission

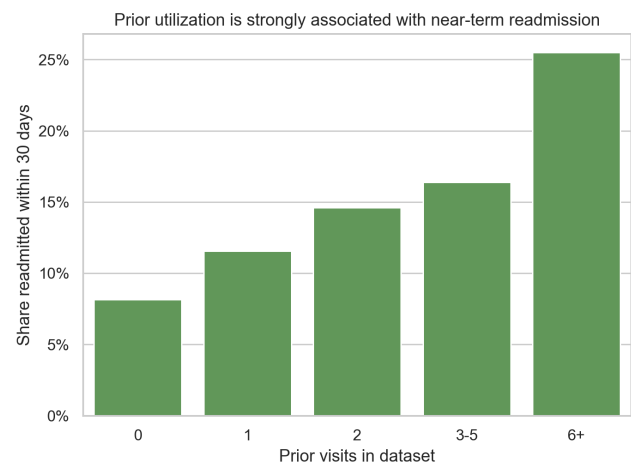


Figure 5: Prior utilization and 30-day readmission

Research question 1: Which patient groups should receive higher follow-up priority? The age plot shows a positive but moderate relationship: 30-day readmission rises from young adulthood into older age and then levels off among the oldest groups. This is consistent with my prior expectation that older patients face more complex diabetes care, but the relationship is not strong enough to justify age-only targeting; it should be used together with other risk markers.

Research question 1: Does prior utilization identify unstable patients? The prior-utilization plot shows the strongest and most monotonic relationship in this section: patients with more previous outpatient, emergency, or inpatient visits have clearly higher 30-day readmission rates. Because the increase is consistent across utilization bands and matches the prior expectation, prior utilization is a practical candidate for discharge-planning triage.

Research question 2: Are A1C testing and glycemic control related to readmission? The A1C plot shows a weaker and less direct relationship than expected: recorded A1C categories differ somewhat in readmission risk, but the “not measured” group remains large and heterogeneous. This suggests that A1C status partly reflects clinical workflow and case mix rather than a simple risk gradient, so I would not interpret the plot as evidence that A1C testing alone identifies the highest-risk subgroup.

4. Dig for the Non-Obvious

Two surprising findings

Finding 1: Medication changes are associated with higher, not lower, 30-day readmission rates.

This matters because a non-analyst might assume that changing diabetes medication during the stay should reduce readmissions. The more plausible interpretation is confounding by indication: clinicians change medication

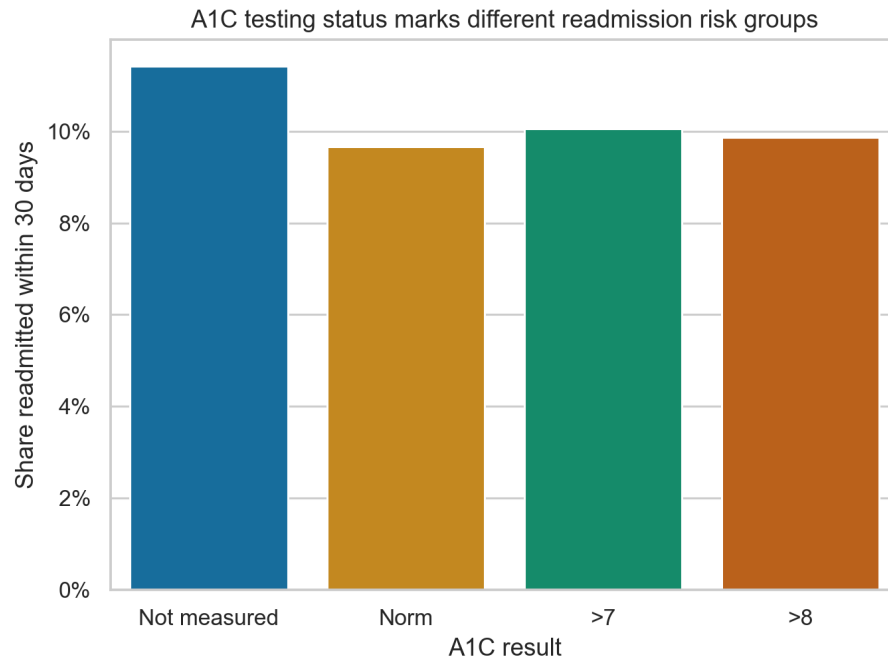


Figure 6: A1C testing status and 30-day readmission

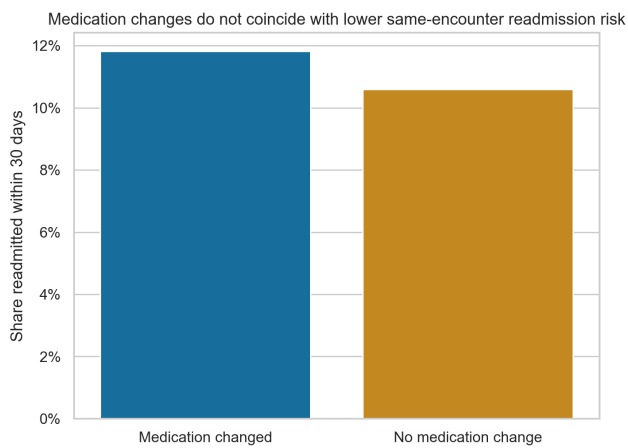


Figure 7: Medication change and readmission

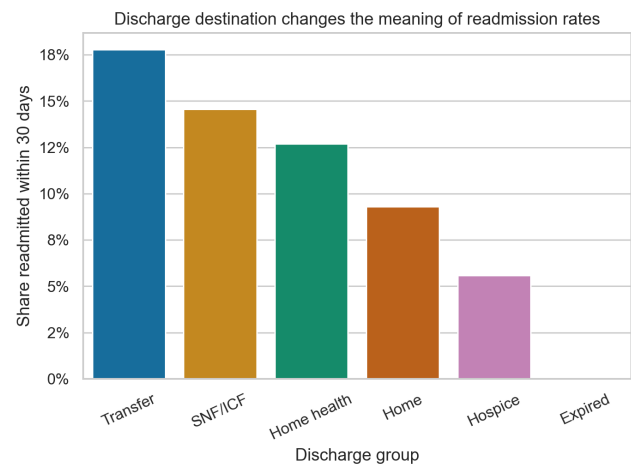


Figure 8: Discharge group and readmission

for patients who are already more unstable or whose diabetes is harder to control. An alternative explanation is documentation: hospitals that record medication changes more carefully may also capture more complex encounters.

Medication-change table, where the mean is the percent readmitted within 30 days:

changed_med	mean	size
Medication changed	11.8	47011
No medication change	10.6	54755

Finding 2: Discharge destination changes the meaning of readmission rates.

Patients discharged to home health, skilled nursing, or transfer settings often have higher readmission than patients discharged directly home. The finding matters because discharge disposition is available at the exact moment when follow-up resources are allocated. However, it should not be read as proof that those destinations cause readmission; they are likely markers of frailty, functional limitations, and clinician concern.

Potential Simpson's paradox or subgroup heterogeneity

The aggregate view suggests that encounters with diabetes medication prescribed have a higher 30-day readmission rate than encounters without diabetes medication. When stratified by age, the relationship is more nuanced:

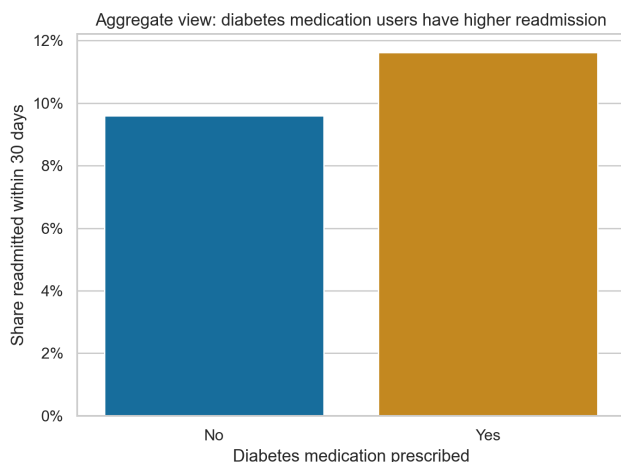


Figure 9: Aggregate diabetes medication relationship

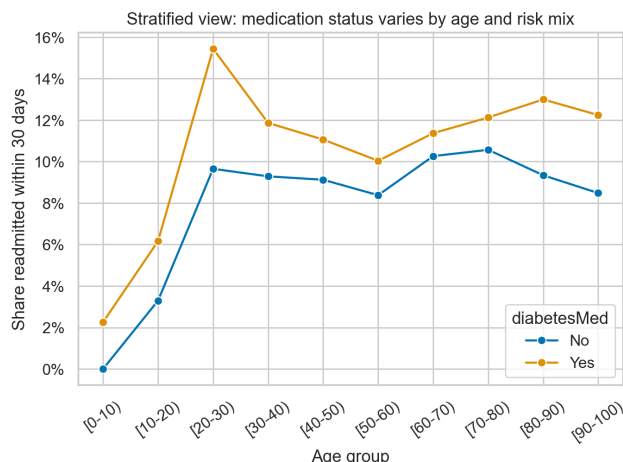


Figure 10: Stratified diabetes medication relationship

This is not a perfect Simpson's paradox because the aggregate relationship does not fully reverse in every subgroup. It is still meaningful heterogeneity: medication status is entangled with age and illness mix, so the aggregate bar chart exaggerates the impression that diabetes medication itself is the problem. The lesson is that an analyst who only looked at the aggregate could mistake a treatment marker for a treatment effect.

Age-stratified table, where the mean is the percent readmitted within 30 days:

age	diabetesMed	mean	size
[0-10)	No	0	28
[0-10)	Yes	2.3	133
[10-20)	No	3.3	91
[10-20)	Yes	6.2	600
[20-30)	No	9.6	342
[20-30)	Yes	15.4	1315
[30-40)	No	9.3	926

age	diabetesMed	mean	size
[30-40)	Yes	11.9	2849
[40-50)	No	9.1	2280
[40-50)	Yes	11.1	7405
[50-60)	No	8.4	3855
[50-60)	Yes	10	13401
[60-70)	No	10.3	4872
[60-70)	Yes	11.4	17611
[70-80)	No	10.6	5877
[70-80)	Yes	12.1	20191
[80-90)	No	9.3	4283
[80-90)	Yes	13	12914
[90-100)	No	8.5	849
[90-100)	Yes	12.2	1944

The Anscombe check

I examined `time_in_hospital` and `num_medications`. Their Pearson correlation is 0.47, suggesting a moderate positive relationship. The four views of this pair reveal progressively more nuance. The raw scatterplot shows that the correlation is real but noisy: at any given length of stay, medication counts span nearly the full observed range. The boxplot makes clear that while the median rises with each additional day, the interquartile spread remains large throughout, meaning length of stay alone is a poor predictor of how many medications a patient receives. Stratifying by readmission outcome shows that the stay–medication relationship is nearly identical across the three readmission categories, so the correlation carries no additional signal about who returns within 30 days. Finally, removing the top 1% of values barely changes the correlation from 0.47 to 0.46, confirming that no small cluster of extreme cases is driving the relationship — the moderate association is genuine and diffuse.

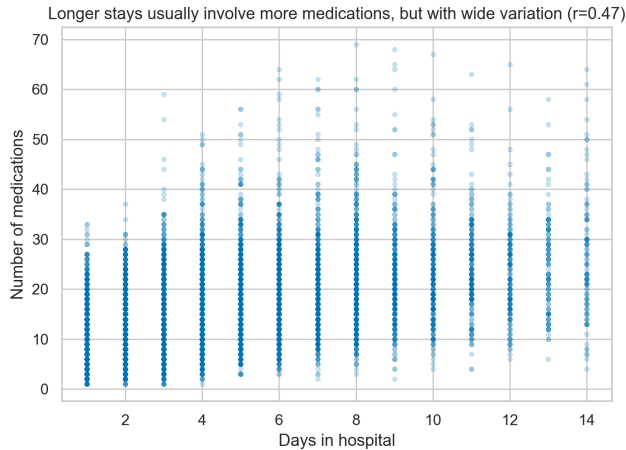


Figure 11: Scatterplot of stay length and medications

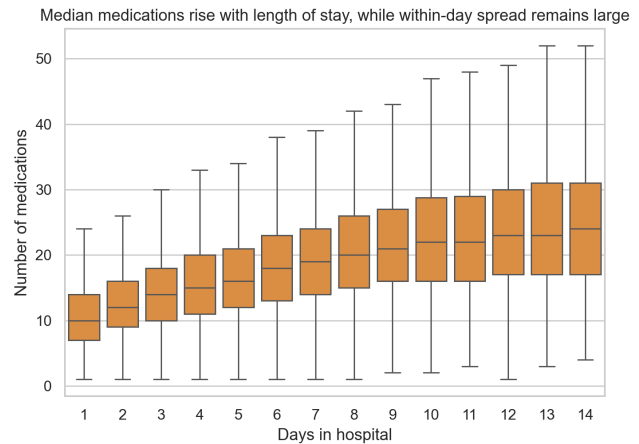


Figure 12: Boxplot of stay length and medications

The full scatterplot shows that longer stays usually involve more medications, but the cloud is very wide: patients with the same length of stay can have very different medication counts. The boxplot makes the central tendency clearer and avoids overplotting, showing a rising median with each additional day. Stratifying by readmission outcome does not reveal a dramatically different shape, which means the correlation should not be reported as if it separates readmitted from non-readmitted patients. After removing the most extreme 1% of values, the correlation remains 0.46, so the relationship is not just an artifact of a few outliers.

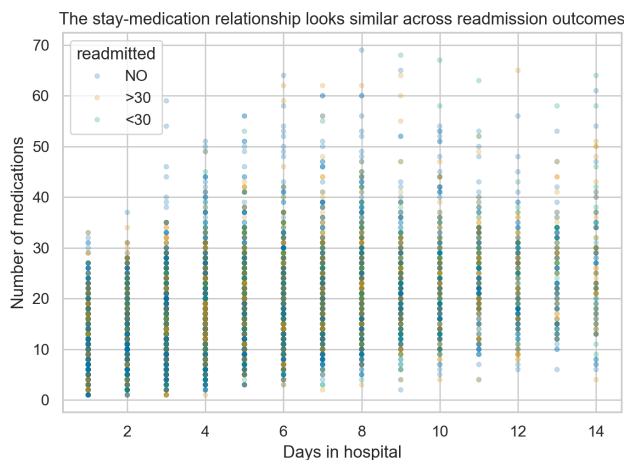


Figure 13: Stratified scatterplot by readmission category

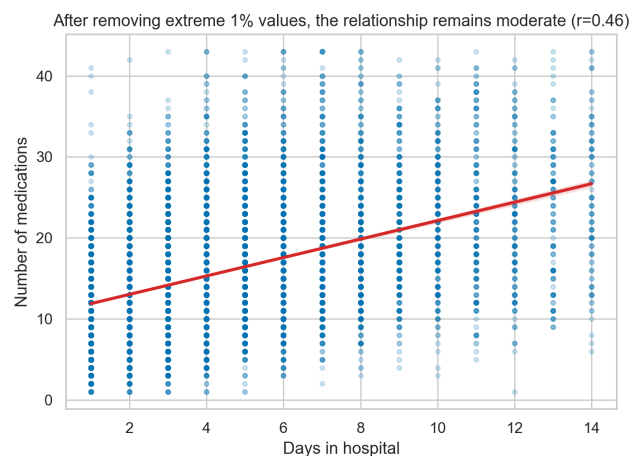


Figure 14: Capped scatterplot

5. Hypotheses and Next Steps

1. **I hypothesize that patients with repeated prior utilization have higher 30-day readmission risk because the EDA shows a steep increase in readmission across prior-visit categories, but this could also be explained by unobserved disease severity or poor outpatient access.** I would test this with a logistic regression or survival model that adjusts for age, discharge destination, number of diagnoses, admission type, and patient-level clustering.
2. **I hypothesize that medication changes during the encounter identify clinically unstable patients rather than causing readmission, because changed-medication encounters have higher readmission rates even though the change was presumably intended to help, but this could also be explained by hospital documentation differences.** I would test this with a propensity-adjusted model or matched comparison using A1C status, medication class, diagnoses, prior utilization, and length of stay.

What data I wish I had

The missing data source I would most want is a post-discharge follow-up record: whether the patient received a primary-care appointment, endocrinology visit, pharmacy reconciliation, nurse call, home-health visit, or transportation support within seven days of discharge. This would directly answer whether readmissions are related to what happens after the hospital stay rather than only to what happened during it. Without it, the current dataset can identify patients who look clinically complex, but it cannot distinguish avoidable readmissions from returns that occurred despite appropriate follow-up. That limits the quality of recommendations because the hospital needs to know which operational actions actually close the gap.

One thing I would tell the client

The clearest action signal is prior utilization: patients who have already cycled through outpatient, emergency, or inpatient care are much more likely to return soon after discharge. I would not use this dataset alone to blame a medication, a discharge destination, or a department, because those fields often mark patients who were already harder to stabilize. My recommendation is to create a discharge checklist that automatically flags diabetic patients with recent prior visits for a follow-up call, medication review, and scheduled appointment before they leave the hospital, then measure whether that process reduces returns over the next quarter. The most important follow-up question for your team is: how many of the patients who were readmitted within 30 days in the past year left the hospital without any of those three touchpoints in place?

6. Reflection

1. What did you do with missing data?

I converted the dataset's ? entries into missing values and kept missingness visible rather than silently filling it. For variables central to the EDA, I either used complete cases for the specific plot or treated missingness as a meaningful category when absence of a test was itself informative, as with A1C. The most problematic variable is weight, which is missing for nearly all encounters; I believe it is MNAR because hospitals likely recorded weight only when it was clinically or administratively salient, not at random. Because of that, I did not use weight in the substantive analysis.

2. Did you remove any observations?

I did not remove observations from the main analysis, because the assignment is exploratory and the unusual cases are part of the hospital's operational reality. For visualization only, I sampled points in dense scatterplots and created one capped view of time_in_hospital and num_medications to check whether the relationship depended on extreme values. I did not remove expired or hospice discharges from the general tables, but I interpreted them carefully because readmission has a different meaning when a patient dies or leaves for end-of-life care.

3. Where did you choose not to explore further?

I deliberately did not go deep into individual ICD diagnosis codes. They are potentially very important, but the dataset stores them as mixed numeric, V-code, and E-code strings, and a responsible analysis would require grouping them into clinically meaningful categories such as circulatory, respiratory, diabetes complications, injury, or neoplasm. That is a worthwhile next step, but it would have shifted the assignment from EDA and problem framing into a medical coding project.

4. What is the single biggest threat to validity?

The biggest threat is that the dataset is encounter-level and observational, while many conclusions the client cares about are patient-level and causal. A field such as medication change, discharge to home health, or long length of stay may look related to readmission because it marks patients who were already sicker, not because it caused the readmission. Repeated encounters from the same patient can also make high-risk patients count multiple times. Therefore, the EDA is strongest as a triage map and weakest as evidence about which intervention will reduce readmissions.